

Data Cleaning and Preprocessing in Excel

Objective

To clean and prepare raw data in Excel by handling missing values, removing duplicates, standardizing text formats, converting date formats, renaming columns, and correcting data types.

1. Identify and Handle Missing Values

Steps: Select the data range. Press Ctrl + G → Special → Blanks → OK. Type a replacement value such as 0, N/A, or Unknown. Press Ctrl + Enter to fill all blank cells simultaneously.

Using Power Query: Data → Get Data → From Table/Range. Right-click the desired column and select Replace Values to replace null values.

2. Remove Duplicate Rows

Select the dataset, go to the Data tab, click Remove Duplicates, choose columns, and click OK. Duplicate records are removed to improve accuracy.

3. Standardize Text Values

Using Flash Fill: Enter corrected values and allow Excel to detect the pattern. Using Find and Replace (Ctrl + H): Replace inconsistent values with standardized values.

Examples: US → United States, M → Male, F → Female.

4. Convert Date Formats

Select the date column, press Ctrl + 1, choose Date, and select the desired format such as DD-MM-YYYY.

If dates are stored as text, use Data → Text to Columns and select the appropriate date format.